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1. General instructions

AT

AT	
describe	Test whether the AT framework is working properly.
response	OK AT command test successful
	ERR AT command test failed
Example	Send: AT
	Response: OK

AT+SAVE

AT+SAVE	
describe	Save Configuration
response	OK Saved successfully
	ERR Save failed
Example	Send: AT+SAVE
	Response: OK

AT+GETVER

AT+GETVER	
describe	Get the software version number
response	OK AT command processed successfully
	ERR AT command processing failed

Example	Send: AT+RESTORE response: getver software:V1.0.0 OK
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AT+RESTART

AT+RESTART	
describe	Reset and restart
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+RESTART Response: OK

AT+RESTORE

AT+RESTORE	
describe	Restore to factory settings
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+RESTORE Response: OK

AT+GETWORKMODE

AT+GETWORKMODE	
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describe	Query working mode (0: Normal working mode 1: Production test mode)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+GETWORKMODE Response: workmode: 0 OK

AT+SETWORKMODE

AT+SETWORKMODE=X	
describe	Set the working mode X: 0: Normal working mode 1: Production test mode
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+SETWORKMODE=1 Response: workmode: 1 ok

2. Development board production test instructions

AT+GETSENSOR

AT+GETSENSOR	
describe	Get acceleration sensor data (Note: BU03 supports, BU04 hardware does not support)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+GETSENSOR

	response: acc_x:0.000000 acc_x:0.000000 acc_x:0.000000 angle:0.000000 OK
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AT+TESTED

AT+TESTLED = X	
describe	Start/stop development board LED (flowing light) test (Note: Use before executing AT+SETCFG to configure device information) X: 1: Start testing LED 0: Stop testing LED
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+TESTLED = 1 Response: OK

AT+TESTOLED

AT+TESTOLED	
describe	Test development board screen (screen display) (Note: Use before executing AT+SETCFG to configure device information)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+TESTOLED=HELLO TEST! Response: OK

AT+DISTANCE

AT+DISTANCE	
describe	Distance measurement (for use after configuration)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+DISTANCE Response: distance: 0.340000 OK

3. TWR algorithm-specific instructions

AT+GETCFG

AT+GETCFG	
describe	Get configuration information
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+GETCFG response: getcfg ID:65535, Role:0, CH:1, Rate:1 OK

AT+SETCFG

AT+SETCFG=X1,X2,X3,X4

describe	Set configuration information (execute the save command after configuration) x1: device ID (0~10) x2: device role (0: tag 1: base station) x3: device channel (0: channel 9 1: channel 5) x4: device rate (0:850K 1:6.8M)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+SETCFG=0,1,0,1 response: setcfg ID:0, Role:1, CH:0, Rate:1 OK

AT+GETDEV

AT+GETDEV	
describe	Get the coefficient of the setting
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+GETDEV Response: OK

AT+SETDEV

AT+SETDEV=X1,X2,X3,X4,X5,X6,X7,X8,X9	
describe	Set the coefficient of the device (execute the save command after configuration) x1: Tag capacity (tag refresh rate) x2: Antenna delay parameters x3: Whether Kalman filter is enabled x4: Kalman filter parameter Q

	x5: Kalman filter parameter R x6: Correction parameter a x7: Correction parameter b x8: Whether to enable the positioning bit x9: Positioning dimension setting
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+SETDEV=5,16336,1,0.018,0.642,1.0000,0.00,0,0 response: <pre>gtdev cap:5 anndelay:16336, kalman_enable:1, kalman_Q:0.018, kalman_R:0.642, para_a:1.0000, para_b:0.00, pos_enable:0, pos_dimen:0 OK</pre>

4. PDOA algorithm-specific instructions

AT+DECA\$

AT+DECA\$	
describe	Pdoa host computer authentication
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+DECA\$ <pre>[14:16:49.521]收←◆J008D{"Info":{"Device":"PDOA Node", "Version":"V1.0.0", "Build":"Jul 19 2024 13:59:55", "Driver":"DW3000 CO Device Driver Version 04.00.00"}}</pre> response: OK

AT+GETDLIST

AT+GETDLIST	
describe	Get [discovery list] (Note: only applicable to base stations)

response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+GETDLIST response: [14:50:48.124]收←◆{DList":["000000004E818834"]} OK

AT+GETKLIST

AT+GETKLIST	
describe	Get [pairing list] (Note: only applicable to base stations)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+GETKLIST response: [14:52:17.056]收←◆{KList:[{"slot":"0001","a64":"000000004E818834","a16":"8834","F":"0001","S":"0040","M":"0000"}]} OK

AT+ADDTAG

AT+ADDTAG	
describe	Add tags to [Pairing List] (Note: only applicable to base stations)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+ADDTAG=000000004E818834,8834,1,64,0 response: [14:51:27.383]收←◆J0064{"TagAdded":{"slot":"0001","a64":"000000004E818834","a16":"8834","F":"0001","S":"0040","M":"0000"}} OK

AT+JOIN

AT+JOIN	
describe	Delete a tag from the [Pairing List] (Note: only applicable to base stations)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+DELTAG=000000004E818834 [14:52:49.883]收到◆J0022{"TagDeleted": "4e818834"} OK response:

AT+PDOAOff

AT+PDOAOff	
describe	Angle correction (Note: only applicable to base stations)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+PDOAOff=1 Response: OK

AT+RNGOff

AT+RNGOff	
describe	Distance correction (Note: only applicable to base stations)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+RNGOff Response: OK

AT+FILTER

AT+FILTER	
describe	Enable filtering (Note: only applicable to base stations)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+FILTER=1 Response: OK

AT+UARTRATE

AT+UARTRATE	
describe	Serial port rate (Note: only applicable to base stations)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+UARTRATE=100 Response: OK

AT+USER_CMD

AT+USER_CMD	
describe	Output format (Note: only applicable to base stations)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+USER_CMD=0 Response: OK

AT+PDOASETCFG

AT+PDOASETCFG	
describe	Set configuration parameters (Note: only applicable to base stations)
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+PDOASETCFG=1,1,3333,1,100,0,0 response: [14:56:25.751]收到←◆gtcfg Dlist:1 KList:1 Net:0D05 AncID:1 Rate:100 Filter:0 UserCmd:0 pdoaOffset:1 rngOffset:0 OK

AT+PDOAGETCFG

AT+PDOAGETCFG	
describe	Get configuration parameters
response	OK AT command processed successfully
	ERR AT command processing failed
Example	Send: AT+PDOAGETCFG response: [14:56:59.940]收到←◆gtcfg Dlist:1 KList:1 Net:0D05 AncID:1 Rate:100 Filter:0 UserCmd:0 pdoaOffset:1 rngOffset:0 OK

5. Algorithm switching instructions

AT+SETUWBMODE

AT+SETUWBMODE=X	
describe	Algorithm selection (save after setting) Xy0yTOWER

	1:PDOA
response	OK AT command processed successfully
	ERR AT command processing failed
Example	send: response:

AT+GETUWBMODE

AT+GETUWBMODE	
describe	Query the current algorithm
response	OK AT command processed successfully
	ERR AT command processing failed
Example	send: response: